Protein Biochemistry
9-8-2017

1. Examine the following peptide sequence:

LKRFDHYELAGREHKI

At pH = 6.0, what is the overall numerical charge of this peptide? Explain your rationale for your answer in detail? (25 pts.)

2. The titration curve of alanine shows the ionization of two functional groups with pKa values of 2.34 and 9.69, corresponding to the ionization of the carboxyl and the protonated amino groups, respectively. The titration of di-, tri-, and larger oligopeptides of alanine also shows the ionization of only two functional groups, although the experimental pKa values are different.

Amino Acid or Peptide	pK1	pK2
Ala	2.34	9.69
Ala-Ala	3.12	8.30
Ala-Ala-Ala	3.39	8.03
Ala-$(Ala)_n$-Ala, n>4	3.42	7.94

A) Draw the structure of Ala-Ala-Ala. Identify the functional groups associated with pK1 and pK2. (20 pts.)

B) Explain why does the value of pK1 increase with each addition of an Ala residue to the Ala oligopeptide? (20 pts.)

C) Provide a rationale for the finding that the value of pK2 decreases with each addition of an Ala residue to the Ala oligopeptide. (20 pts.)

3. At the pI (Isoelectric point) of a biomacromolecule, such as a protein, one finds that this is where the biomacromolecule displays its lowest levels of solubility. Explain in detail this finding. (15 pts.)